Boltzmann Statistics

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1 Introduction

To move on to polymer configuration, we should understand Boltzmann statistics. So, to derive them, we unfortunately need to pull another somewhat unexplained concept out of the hat: entropy. From the laws of thermodynamics, we learn that for a system in equilibrium, entropy will be maximized. This means we are heading towards maximal disorder, and this idea is responsible for the eventual heat death of the universe, where everything will be come too disordered for much interesting structure to exist. However, here we will use it to derive one of the foundational equations of statistical mechanics. Generally we see entropy written as such:

$$S = k_B ln(W)$$

Here, S is entropy, and W is the number of microstates a system has. This is a formula you will just have to take at face value, but I will try to justify it a little bit.

1.1 Justifying entropy

First, let's talk about microstates. The classical example is that of flipping 2 coins, where there are 4 microstates and 3 macrostates. A macrostate is some total configuration of a system, for example, 2 heads is a macrostate. A given macrostate may have multiple different ways to make it though, and we call those microstates. For example, below, for the macrostate of having 1 coin land heads and 1 tails, there are 2 unique microstates that make that macrostate.

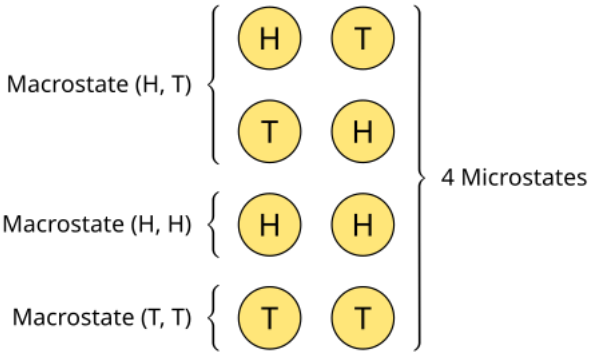


Figure 1: An example of one random walk path of a polymer

To connect this with random walks, the macrostate could be the final position of the walk, but as you can imagine there are many different ways the random walk can go to get to a certain final position. So, because a random walk (of large enough length) has more microstates that this 2 coin system, it would have a higher entropy.

But what is entropy? Generally it is said to be a measure of disorder. Because of this, it should have a key property:

$$S(A + B) = S(A) + S(B)$$

This says that if you have 2 independent systems and measure their entropies individually and together, you will get the same total result. Microstates could be a good measure of disorder, so let's think about doing that. However,

what you would find is that using microstates directly does not work. This is because if you have a system A with microstates W_A and a system B with microstates W_B, what you find is that $W_{A+B} = W_A * W_B$. If we just say entropy is proportional to microstates, this would violate the property that we want to satisfy. It turns out that to satisfy this property, we can just take the log of the number of microstates, and then the property is satisfied.

1.2 Questions

1. Try testing $W_{A+B} = W_A * W_B$ by considering a system of 2 coins and a system of 1 coin and convince yourself it is true. Can you see that this works for random walks too?

2. Using log properties show that if $W_{A+B} = W_A * W_B$ then $S(A + B) = S(A) + S(B)$ if you use the proper formula for entropy.

2 Deriving the Shannon entropy

To take a step towards the Boltzmann statistics formula, next we must find a different expression for entropy. Let's think about this in the context of a 1-d random walk, where we have N total steps, of which n_l are to the left and n_r are to the right. We can think of this as arranging all the steps in a random order. Consider each individual step, and think about it as going in a random spot in the random walk. The first step you put in has N different options. The second will have $N - 1$ (Since one is already occupied,) and so you will find that there are $N!$ different configurations.

However, there is one subtlety. Each step that goes in the same direction (so say all the left steps) are, other than their position in the walk, indistinguishable from each other. This may seem weird, but nature says that if you have particles (or steps in this case) that are indistinguishable from each other, you only get one microstate for each total arrangement. If we go back to the coin example, if you can't tell the coins apart, this means there are actually only 3 microstates, because the two states where one is heads and one is tails actually just become one. Thus, the number of microstates for this 1-d random walk becomes $\frac{N!}{n_l! n_r!}$. We can now write the entropy as such:

$$S = k_B ln(\frac{N!}{n_l! n_r!}) = k_B[ln(N!) - ln(n_1!) - ln(n_2!)$$

There are 3 properties we must invoke to make progress.

1. By definition,
$$N = n_l + n_r$$

2. We can write the number of states as probabilities as such:
$$p_l = \frac{n_l}{N} \qquad\qquad p_r = \frac{n_r}{N}$$

3. Finally, there is a property of taking logs of factorials that is very useful here, called Stirling's approximation, which says that

$$ln(N!) \approx Nln(N) - N$$

for a big enough N. In this case, we will assume all of N, n_l and n_r are big enough to use this.

2.1 Derivation time

Staring from the entropy and using the above properties, try to write the entropy purely in terms of probabilities and N. Hint: start by using Stirling's approximation on all the ln terms in the entropy, after which there will be some terms that cancel. Next, plug in for all n_ls and n_rs with probabilities and Ns, and then expand the logs again. There will be some more terms which cancel, and if you want to confirm your answer, there is an extrapolation of this answer to a more general formula further below.

Ok, from what you derived, you may be able to see what the general form should be. We write Shannon entropy as

$$S = -Nk_B \sum_i p_i ln(p_i)$$

The next step is to maximize this quantity, and see what happens.

3 Maximizing the Shannon entropy

Now, to maximize the entropy, all we need to do is take a derivative of S with respect to p_i, and then set it to 0, right? Well, let's see what happens.

$$\frac{\partial}{\partial p_i}[-Nk_B \sum_i p_i ln(p_i)] = 0$$

$$-Nk_B \frac{\partial}{\partial p_i}[p_i ln(p_i)] = 0$$

$$ln(p_i) + \frac{p_i}{p_i} = 0$$

$$ln(p_i) = -1$$

$$p_i = \frac{1}{e}$$

But this can't be correct. No matter what the state is, its probability is the same? And on top of that, this violates the laws of probability, because the probabilities won't add up to 1! And where is the energy and temperature dependence?

To tackle this problem, we need to introduce some constraints that we know to be true, but aren't enforced in the math we are doing. In this case, there are two:

1. The probabilities must add up to 1.
$$\sum_i p_i = 1$$

2. The energies of states times their respective probabilities must give the average energy.
$$\sum_i p_i E_i =< E >$$

So how do we introduce these constraints mathematically? We will use something called Lagrange multipliers, which is just a fancy way of saying we will add 0 into what we are minimizing. Consider rewriting the constraints above as $(\sum_i p_i) - 1 = 0$ and $(\sum_i p_i E_i) - < E >= 0$. Both of these constraints are now shown as something being equal to 0. If we then multiply them by a number (this number is called the Lagrange multiplier) this is still true:

$$\lambda((\sum_i p_i) - 1) = 0$$

$$\beta((\sum_i p_i E_i) - < E >) = 0$$

Here, λ and β are the Lagrange multipliers. So now, what we can do, instead of minimizing just the entropy, let's minimize the entropy plus these 2 constraints that we know to be 0. (And because they are 0 their derivatives should be too.) Mathematically this OK: instead of maximizing the entropy, we are maximizing the entropy + 0.

$$\frac{\partial}{\partial p_i} - \sum_i p_i ln(p_i) + \lambda((\sum_i p_i) - 1) + \beta((\sum_i p_i E_i) - < E >)] = 0$$

Go ahead and take this derivative below, and remember that we are only taking the derivative with respect to a single p_i, so all terms in sums except one will vanish. Then solve for p_i. (If you're concerned that the Nk_B vanished, you can just imagine that it disappeared into the β and λ)

The answer you should have found is $p_i = e^{\lambda-1+\beta E_i}$. We can simplify this a bit as such:

$$p_i = e^{\lambda-1+\beta E_i}$$

$$p_i = e^{\lambda-1}e^{\beta E_i}$$

$$p_i = Ce^{\beta E_i} \qquad C = e^{\lambda-1}$$

Wow! We have found the exponential relationship between probability and energy! There is one more thing to fix in the exponential though, which is that we don't know what β is. So, let's do some reasoning to figure it out. There are 3 things we need β to satisfy:

1. We know that the higher the energy of a state, the lower its probability should be. To make that happen, β should be negative.
2. We should incorporate temperature dependence into β as well. We know that at higher temperatures, higher energy states are more likely. Since the exponent is negative, to get this behavior, β should be inversely proportional to temperature, ie. $\beta \propto \frac{1}{T}$
3. Finally, we need to fix the units in the exponent. The argument of an exponent may not have units, so β must have units of one over energy. We already have an inverse temperature dependence, and so we should introduce constants to get us to the correct units. It turns out that k_B times the temperature gives units of energy, so why not use the constant we are already working with, and stick a k_B in the denominator of β.

All together, this gives us $\beta = -\frac{1}{k_B T}$, which gives us

$$p_i = Ce^{-E_i/k_B T}$$

There is just one more thing unknown to us, and that is C. We know it must be positive, since $C = e^{\lambda-1}$, but that doesn't help us too much here. To figure out what it is, let's go back to one of our previous constraints:

$$\sum_i p_i = 1$$

Using this, find C below.

You should have found that C is 1 over a sum. People usually call this sum Z, so that $C = \frac{1}{Z}$. This leads to the final formulation of Boltzmann statistics:

$$P(i) = \frac{e^{-E_i/k_B T}}{Z}$$

where

$$Z = \sum_i e^{-E_i/k_B T}$$